

6.7800 Project Part #1

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1. (a) Let $y = (y_1, \dots, y_N)$. The probability that we get y_1 is $P_{f^{-1}(y_1)}$, and the probability of obtaining y_k after y_{k-1} is $M_{f^{-1}(y_k), f^{-1}(y_{k-1})}$. So,

$$P(y | f) = P_{f^{-1}(y_1)} \prod_{k=2}^N M_{f^{-1}(y_k), f^{-1}(y_{k-1})}.$$

- (b) By Bayes',

$$\begin{aligned} p_{f|y}(f | y) &= \frac{p_{y|f}(y | f) p_f(f)}{p_y(y)} \\ &= \frac{P_{f^{-1}(y_1)} \prod_{k=2}^N M_{f^{-1}(y_k), f^{-1}(y_{k-1})}}{\sum_{g \in S_N} P_{g^{-1}(y_1)} \prod_{k=2}^N M_{g^{-1}(y_k), g^{-1}(y_{k-1})}}. \end{aligned}$$

The MAP estimator of f is the permutation that maximizes the numerator, aka $\hat{f}_{MAP} = \arg \max_f p_{f|y}(f | y) = \arg \max_f p_{y|f}(y | f)$.

- (c) Maximizing $p_{y|f}(y | f)$ is computationally infeasible because there $N!$ factorial different possibilities, and its difficult to come up with an algorithm that can do better.

2. (a) There are $m!^2$ pairs of permutations, and there are $m!$ ways to choose the first permutation f_1 . Then, there are $\binom{m}{2}$ different permutations that differ from f_1 in exactly two positions (aka a transposition of f_1). Thus, the answer is

$$\frac{\binom{m}{2}}{m!} = \frac{1}{2(m-2)!}.$$

- (b) We use the distribution from 2(a) as our proposal distribution. That is,

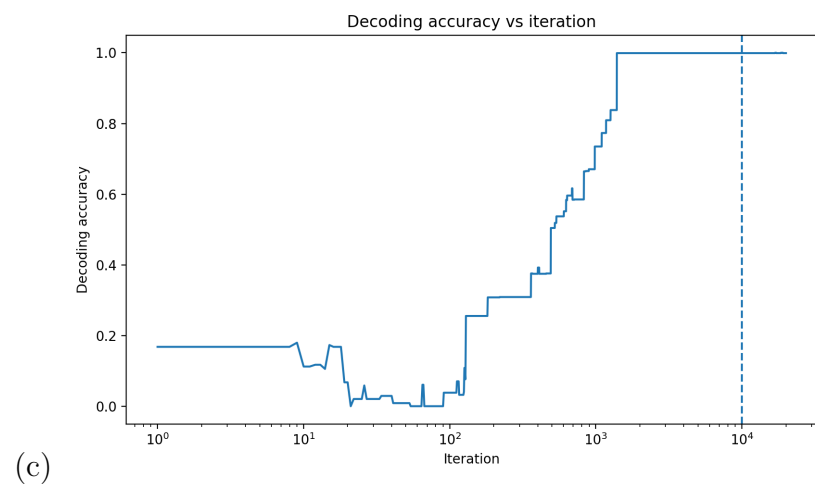
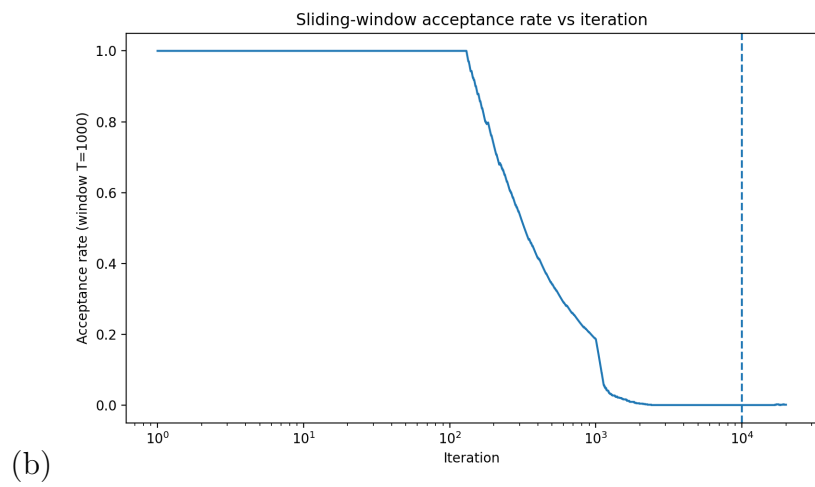
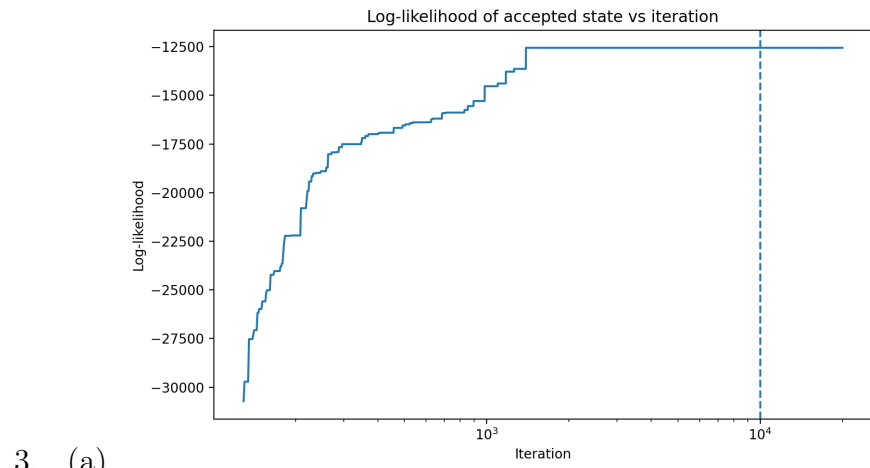
$$v(f' | f) = \frac{1}{2(m-2)!}.$$

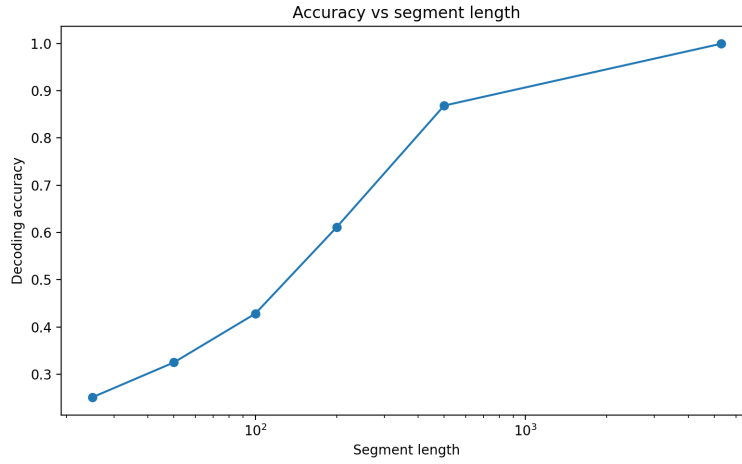
Thus, the MH algorithm produces the following Markov chain:

- i. Choose x_0 arbitrarily from possible $f \in S_n$.
- ii. Assume we have some x_n .
- iii. Generate a new f' from f by using the distribution $v(\cdot | f)$.
- iv. The acceptance factor is

$$a(f \rightarrow f') = \min \left(1, \frac{p_{y|f}(y | f')}{p_{y|f}(y | f)} \right).$$

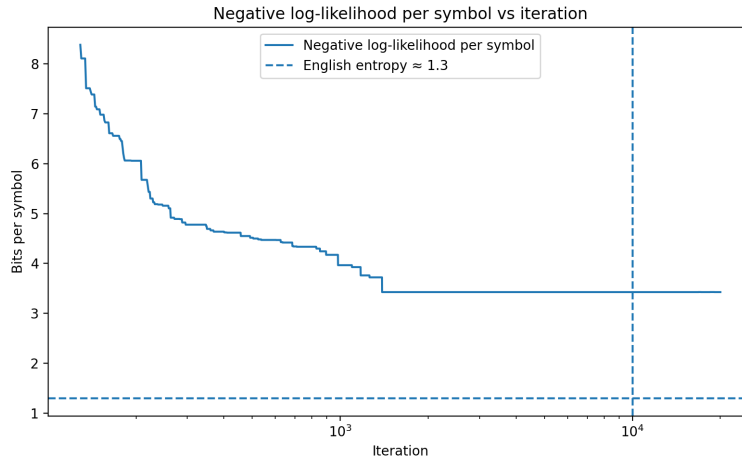
- v. Now choose either to accept or reject this transition with probability $a(f \rightarrow f')$, and this becomes our new x_{n+1} .
- (c)
- i. Sample an f from the a posterior distribution by running the algorithm in 2(b) sufficiently many times.
 - ii. Draw many samples from the posterior distribution.
 - iii. The permutation with the highest frequency is the MAP estimator.





(d)

Longer segments lead to higher accuracy. This is because on shorter segments, there is more noise and less accurate estimations can do decently as well as the optimal, while for longer segments there's more statistical evidence and the optimal does a lot better than things that aren't optimal.



(e)

Because it is only a Markov chain looking at the previous characters, it doesn't cover English semantics or longer range contexts. Therefore entropy is likely to be higher.